

BEAM SYNCHRONOUS FOR THE REST OF US!

A Precision Machine Timing Proposal

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Revision History

Revision	Date	Description of Change
-	08/01/2024	Initial Release, Draft Status
A	09/04/2024	Ready for Review

Acknowledgements

This document utilizes concepts from both the GSI/FAIR and LCLS-II timing system upgrades. It is based heavily on prior work for the PIP-II LCLK timing system as documented by Mark Austin, Dan McArthur, and Greg Vogel. Many thanks to Ed Cullerton and Brian Schupbach for helping with the author's understanding Booster transfer scenarios and LLRF functionality.

1.0 ABSTRACT

This document outlines a method by which local beam-synchronous events may be distributed through asynchronous serial protocols. This method is ideal for synchrotron machines with aggressive frequency sweeps (such as Fermilab's 38~53MHz Booster) and allows for distributed timing to be maintained without specialized hardware. The method assumes both local LLRF and global PPS reference signals are available at distribution nodes in addition to a general-purpose serial link. This document is a proposal and does not encompass the final protocol specifications.

1.1 Executive Summary

This proposal seeks to utilize White Rabbit technology as a drift compensation mechanism for the Accelerator Clock (ACLK) and Beam Synchronous Clock (BSCLK) distribution networks. Incorporation of a global reference signal to synchronized local oscillators will ensure machine timing plays out at an inter-bunch level in a reliable manner. This solution comes at negligible reduction to latency and maintains Fermilab's current real-time control paradigms.

1.2 Background

Fermilab's Tevatron Clock (TCLK) infrastructure has been an integral part of the accelerator control network since the 1980's. This 10MHz Manchester encoded protocol has enabled flexible, real-time event distribution for thousands of devices connected to the timing network with a high degree of reliability.

Forthcoming upgrades to the Fermilab complex (PIP-II, LBNF, ACORN) necessitate higher levels of precision to maintain inter-bunch timing for Instrumentation and Control purposes. This presents as an opportunity to refine the event distribution protocol for tighter synchronization between machines, experiments, and eventually far-site operations.

1.3 Terms Used In This Document (TUITD)

The following terms are defined:

- Accuracy – The deviation of an event edge from where it entered the timing system.
- Clock Cycle – The standard clock 'tic' of a device in the timing system.
- Device – A discrete component of the timing system.
- Event – A singular, distinct point of time to be distributed by the timing system.
- Event Window – The finest resolution between events in the timing system.
- Granularity – The finest resolution between clock cycles in the timing system.
- Grandmaster Clock – Official time device referenced by the timing system
- Jitter – The statistical deviation of an event edge in the timing system.
- Latency – The propagation delay for a signal to traverse the timing system.
- Node – Terminus or consumer of events in the timing system.
- Non-real-time – Actions that may be deferred or loosely defined.
- Offset – The intentional or measured delay of a signal in the timing system.
- Precision – The deviation of event edges between two points or periods in the timing system.
- Phase – Relation of one device's clock cycle to another's in the timing system.
- Phase Delay – The alignment mismatch between a periodic signal measured at two points.
- PPS – Pulse Per Second, a high-accuracy repeating signal marking one second.
- Real time – Actions to be taken 'as fast as possible'.
- Repeater – Signal 'fanout' or 'pass-through' device utilized by the timing system.
- RF Synchronous – An external signal linked to a clock or event in the timing system.
- Signal – The medium through which events are transmitted in the timing system.
- Source – Origination device utilized by the timing system
- Super Cycle – Complete execution of a timeline by the timing system.
- Time – What a clock reads
- Timeline – A sequential series of events executed by the timing system.
- Trigger – An action instigating or instigated by an event.

1.4 Concepts

Two concepts are critical for understanding this proposal. Please review the ‘Concepts’ section for further information regarding the nomenclature used herein.

1.4.1 GLOBAL REFERENCE

Distributed timing systems are subject to both propagation (phase) delay, and signal jitter. For a timing system to act in a beam-synchronous (BS) manner, the phase delay must be held constant relative to the beam at an inter-bunch level. This proposal utilizes a global reference Pulse Per Second signal (PPS, pronounced ‘peepis’) facilitated by White Rabbit (WR) or Global Position System (GPS) hardware at each location to measure this phase delay. This PPS is to arrive to each node simultaneously, once per second, irrespective of propagation, with sub-nanosecond jitter.

1.4.2 EVENT WINDOWS

For this proposal, Event Windows (EW) are utilized to align nodes in the time domain. These event windows happen simultaneously for each node in the timing network and represent the finest granularity at which events may occur. Due to propagation delay or jitter, events may fall into adjacent Event Windows during distribution. These events are *always* re-aligned by the timing system to a subsequent event window in order to maintain integer window delays. For BS discussion purposes, the EW granularity may be considered a subharmonic of the LLRF frequency of the machine (presently 7.5MHz for Main Injector). For general discussion purposes the EW granularity will be considered 10MHz, or 100ns.

1.5 Technical Specifications

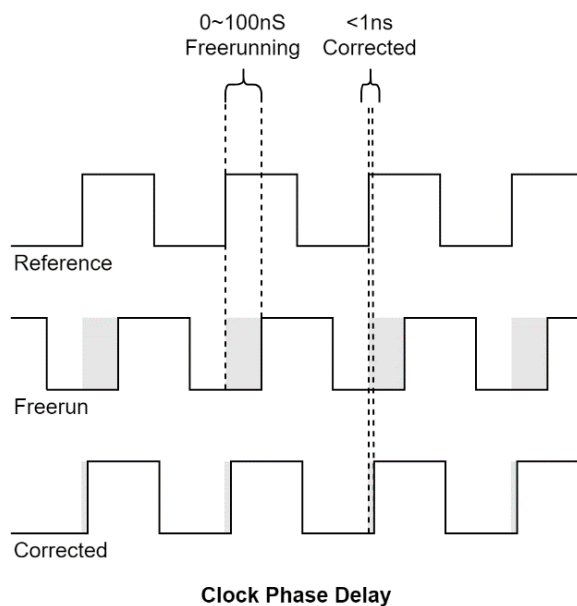
General Machine Timing	Min	Max	Nom	
Base Freq.			10	MHz
Granularity			100	ns
Jitter			<1	ns (WR)
Data Rate		1000	100	Mbps
Event Spacing			1	us
Repeater Latency	10	500	200	ns
Upper Bound Lat.			50	us
Beam-Synchronous	Min	Max	Nom	
Base Frequency	7.5	16.25	10	MHz
Granularity	133	61.5	100	ns
Jitter		±18	<1	ns (WR)
Data Rate	75	1000	100	Mbps
Event Spacing	1.33	0.615	1	us
Repeater Latency	10	500	200	ns
Upper Bound Lat.			50	us

2.0 GENERAL MACHINE TIMING

General Machine Timing (GMT) will be comprised of 100ns ACLK event windows, synchronized by each node to a WR reference oscillator. Events will be distributed in real time through dedicated serial links and aligned locally to the global 10MHz reference. The following principles and technical details are explicitly defined for events to be distributed in the GMT domain.

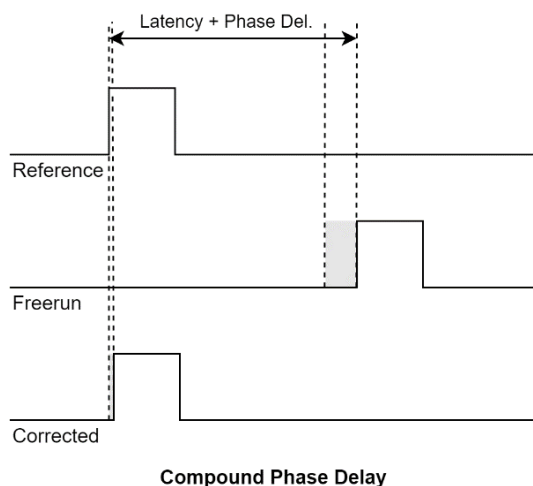
2.1 Clock Phase Delay Correction

One source of jitter between nodes is the phase delay between clocks. While two clocks may share an identical frequency with minimal deviation between clock periods (both clocks oscillate at exactly 10MHz) the deviation between edges may vary according to network latencies or gradual drift of the oscillator. This variance between clock edges is known as 'phase delay' and will impart jitter to the granularity window according to the period of the clock (10MHz $\rightarrow$ 100ns). Clock Phase Delay is corrected via the White Rabbit technology.



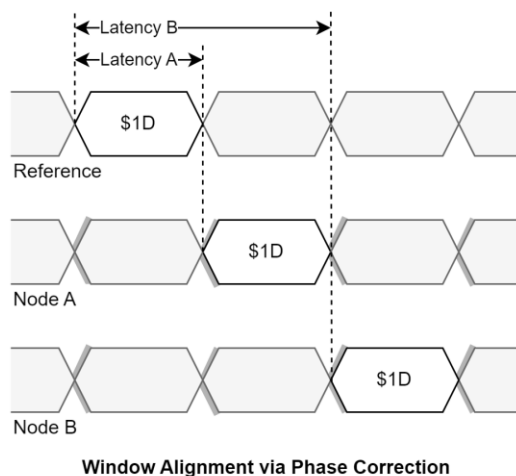
2.2 Compound Phase Delay Correction

The difference of a selected clock cycle two nodes is the compound phase delay. This is similar to event latency but may be corrected for by utilizing a technology such as White Rabbit. In this scenario, a hardware-level protocol is utilized to synchronize the two clocks at 1:1 level. Each device connected to the White Rabbit timing network will receive an identical with Pulse-Per-Second (PPS) with sub nanosecond accuracy. Note! This PPS arrives simultaneously to all devices and is not considered an Event which is subject to propagation delay.



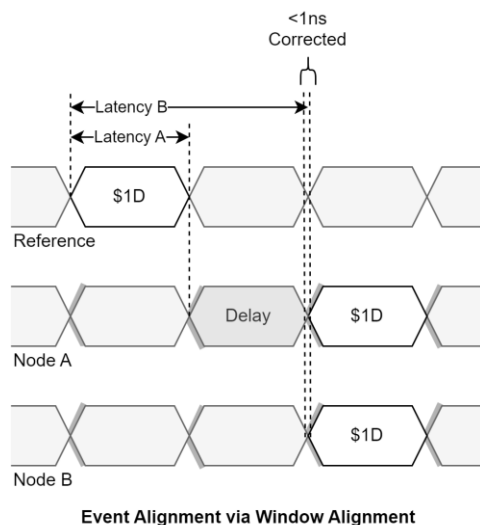
2.3 Window Alignment

Given a series of synchronized clocks connected to a timing network, real-time event windows may be aligned to a high degree of precision. In this scenario, the White Rabbit network holds each local oscillator to the same 10MHz frequency. While events may arrive with different latencies or jitter according to varying network conditions, their actions are coordinated to the same clock edge across the complex, negating the compound error effects outlined above.



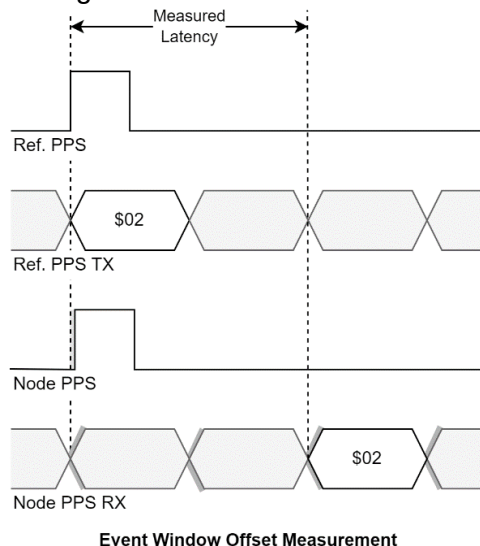
2.4 Event Alignment

Given a series of synchronized event windows for devices connected to a timing network, real-time events may be synchronized to a high degree of precision by applying an event window delay according to the maximum upper bound latency of the furthest node. Utilizing the clock-alignment technology provided by White Rabbit, two distributed real-time events can act at the same time by applying a fixed event window offset as determined by the delta between devices.



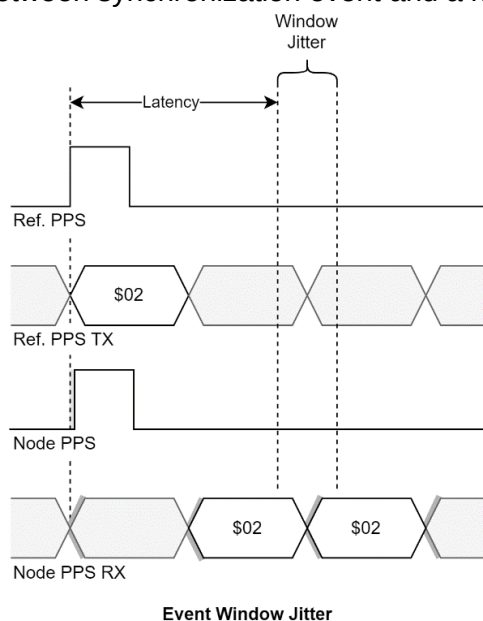
2.5 Event Window Offset Measurement

The event window offset requires a periodic synchronization event to be measured against the PPS signal at both units. This synchronization event determines the accuracy of all connected real-time devices and should be the highest priority event distributed by the timing network. The determination of the event window offset is accomplished by measuring the number of windows to elapse between the local PPS signal and event arrival time as shown below:

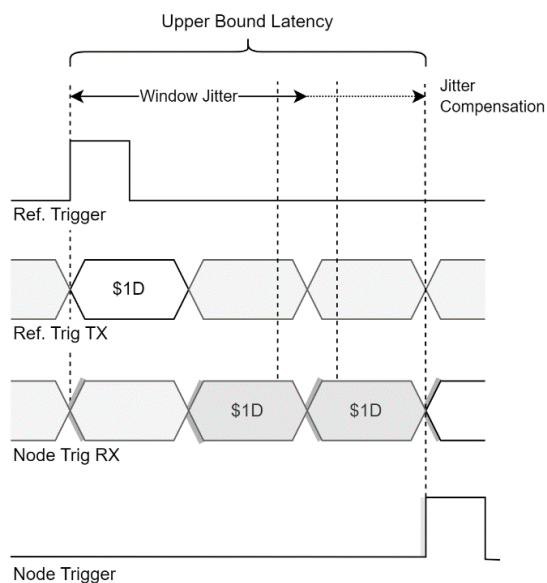


2.6 Event Window Jitter

Variations in network condition may cause synchronization event arrival time to jitter between adjacent event windows. In this scenario, subsequent event actions will vary by the event window period, or 100ns for a 10MHz clock. For this reason, all real-time devices should record the event window offset at every synchronization cycle and utilize a buffer window to negate the effect of event arrival jitter. Stated differently: event actions should align to a 1~3 window delay according to the difference between synchronization event and a nominal PPS signal.



Event Window Jitter



Event Window Jitter Correction

3.0 BEAM SYNCHRONOUS TIMING VIA GMT

The Event Window methodologies outlined above are predicated on a local 10MHz oscillator that operates independently from the communication protocol and its latencies. This timing method, asynchronous to the serial link, can be adapted to trigger from other oscillators operating at arbitrary or sweeping RF frequencies. By appending a 'bucket count' to distributed events, triggers may be transposed to RF event windows with a high degree of precision and repeatability. The GMT system can then be used to control distributed accelerator systems at an inter-bunch level.

3.1 Reset Alignment

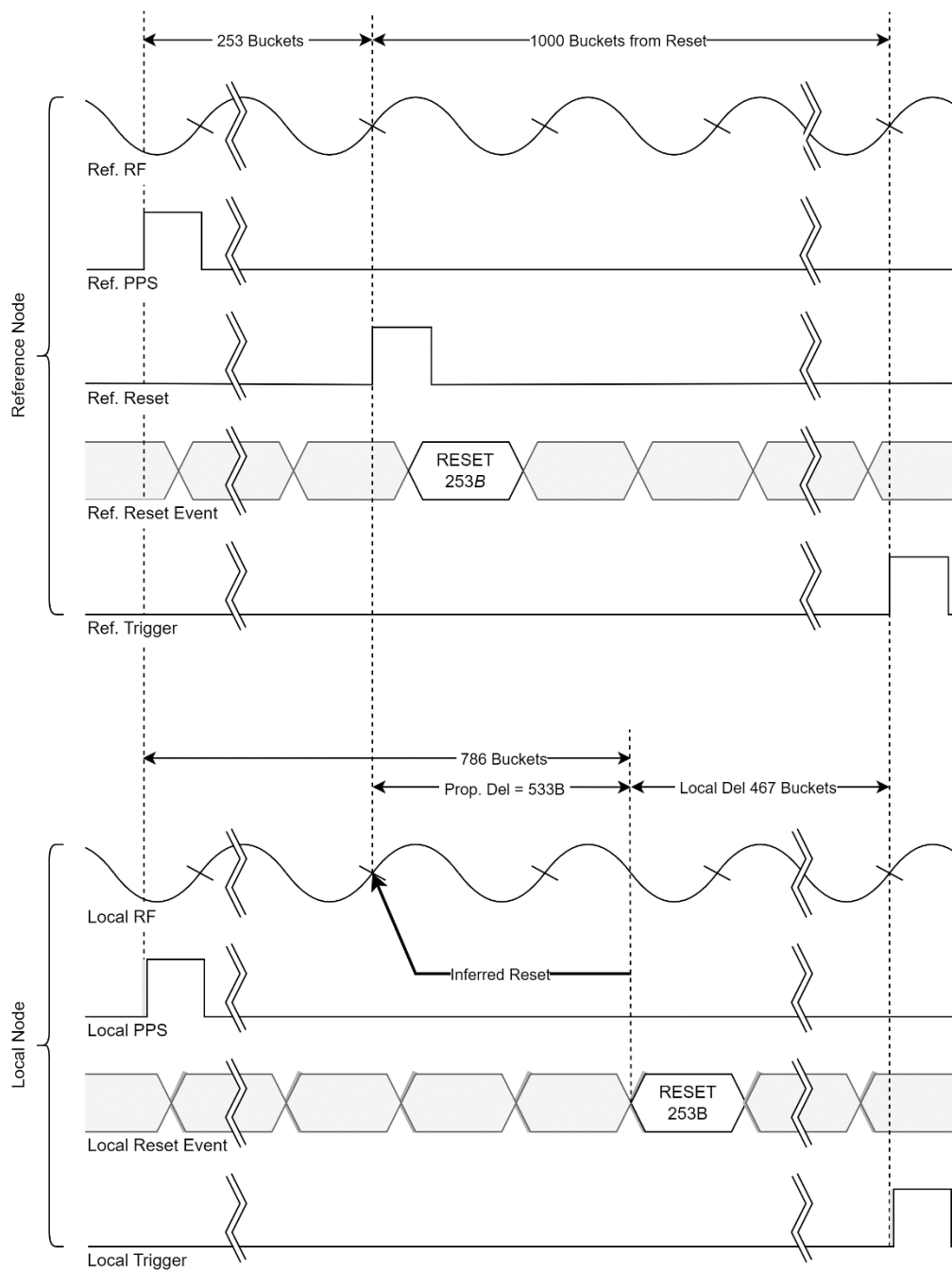
Reset or 'injection' events are a fixed marker, aligned to the first bucket in a bunch train to enter a machine. From this marker and number of RF periods elapsed during the acceleration period, it is possible to track a target bunch throughout the machine cycle. This synchronization is accomplished in the proposed GMT system by measuring the RF zero-crossing events elapsed against the global PPS reference and applying a 'local bucket offset' to events to be triggered at an exact RF cycle.

The scenario operates as follows:

1. Both reference and local timing nodes count RF buckets in reference to the global PPS. The precision of this count will depend on local phase-matching and LLRF distribution. At Fermilab, the LLRF infrastructure is generally considered to be 'good enough.'
2. On machine reset, the reference timing device generates a reset event along with the number of buckets elapsed since PPS and inserts it into the timeline.
3. The local node decodes the event in the static 10MHz event window and utilizes an internal bucket count to infer the bucket propagation delay and originating reset bucket.
4. This RF bucket propagation delay may then be used to align the bunch train, track specific RF cycles, and trigger global inter-bucket events relative to the local RF.

In the example timing diagram below, the machine reset trigger occurs '253' buckets after the PPS signal (an arbitrary number for example purposes). This reset event is broadcast to the general timing network, and received by a node that has measured '786' buckets since its local PPS signal was received. From this, the instantaneous propagation delay is determined to be '553' buckets ($786 - 253 = 553$). For a global event configured to trigger after 1000 RF cycles, the local node waits '467' buckets ($1000 - 553 = 467$) before firing at an LLRF zero crossing.

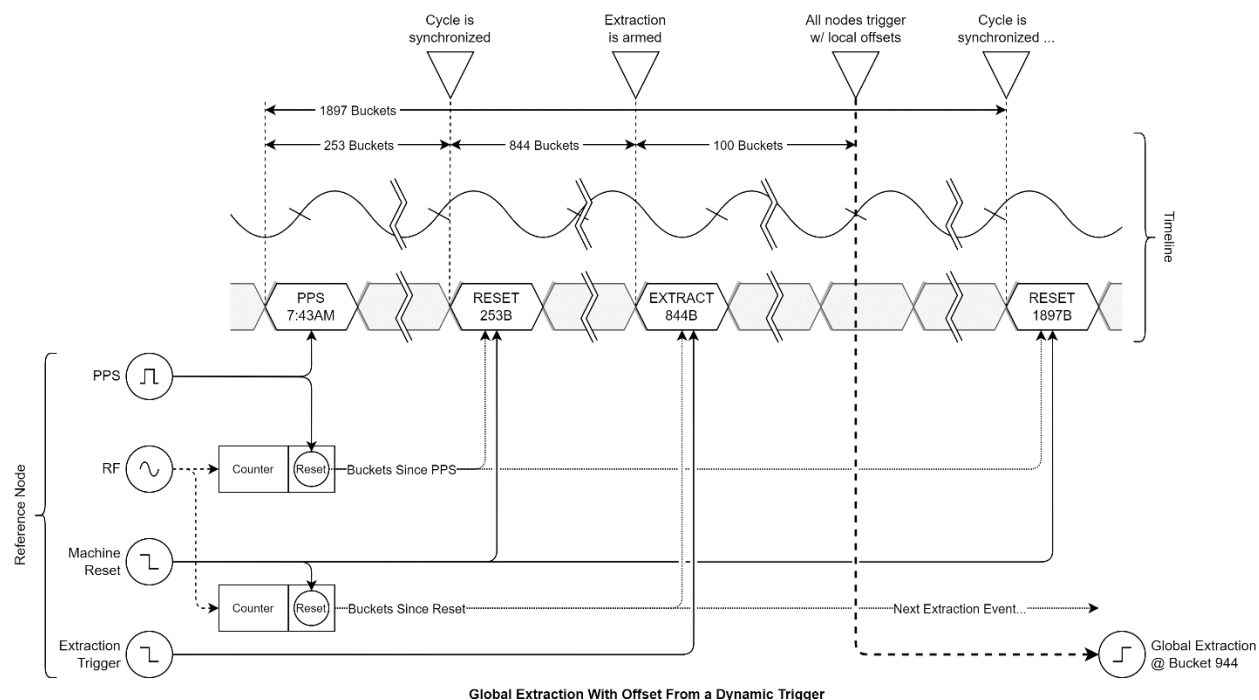
**The reset event may have different meanings according to the context in which it is used. For this document, the reset event is defined as 'event corresponding to the first bunch to enter the machine'*



Inter-Bucket Reset Alignment in Asynchronous Clock Domains

3.2 Inter-Reset Events

To simplify events that occur every machine cycle, GMT events targeting exact buckets may be injected into the timeline with a ‘bucket offset’ relative to the preceding machine reset. In the example below, asynchronous reset and extraction events are measured relative to the reference RF and inserted into the timeline. An external extraction trigger fires 844 buckets after the machine reset. All nodes are programmed to trigger 100 buckets from the extraction event. Local nodes correlate this delay with their individual bucket offset and execute globally on a given bucket, asynchronous of the GMT event windows.

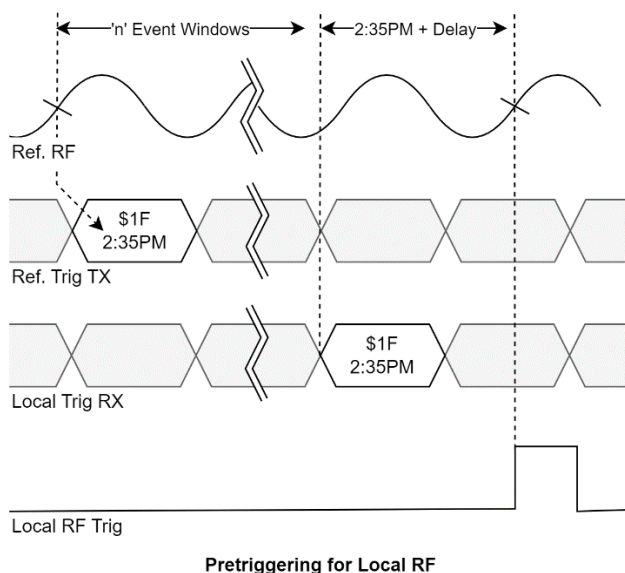


3.3 BS Event Window Jitter

The asynchronous-ness between the global GMT and BS triggers will definitively impact the accuracy of events as they are encoded into the 10MHz clock domain. This is partially addressed by re-aligning the bunch train at a fixed point in every cycle but may still result in a nominal window jitter of $\pm$ one RF period. To compensate for this, an upper-bound latency measurement of the RF propagation delay may be made, and corrections applied in a similar manner to the ‘Event Window Jitter’ correction method stated above. This approach is still under analysis.

3.4 BS Events without LLRF.

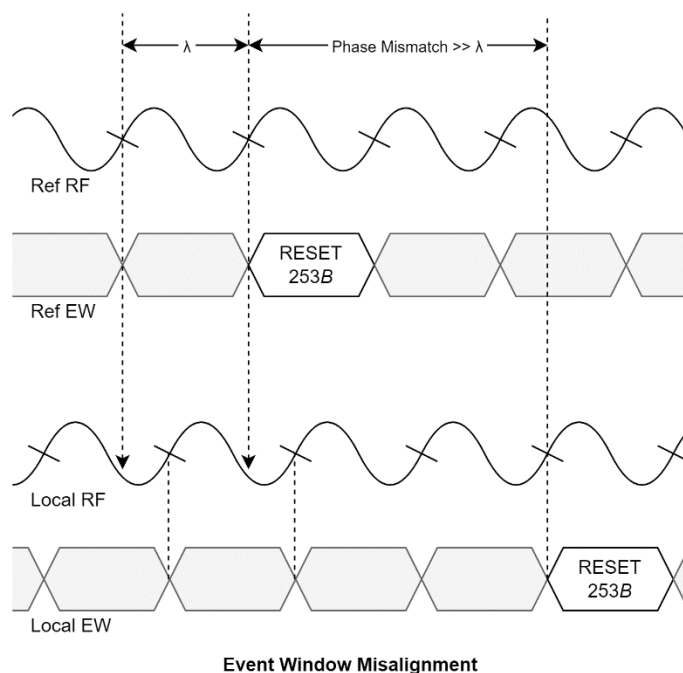
For fixed RF systems (PIP-II) requiring real-time event actions aligned with frequencies anharmonic to the timing system, a measured time delay may be broadcast with the RF synchronous event in lieu of a bucket count. The downstream node may then apply a fixed time delay corresponding with the period of the number of RF buckets elapsed. For two points along a beamline, the RF phase delay between the two will remain fixed on a batch-to-batch basis. Given a fixed-latency timing system, RF events may be reliably measured and acted upon by transmitting and applying this Δt between zero crossings and start of event window as shown:



4.0 DEDICATED BS TIMING NETWORKS

For machine-specific actions to be performed repeatedly at high rates, a dedicated BS timing network may be required. This RF-dependent timing may be performed in a similar manner to the GMT proposal above, but with event windows aligned precisely to the local RF domain. In this scenario, Event Windows are to be maintained constant to the immediate LLRF signal and cannot be 'phase aligned' between nodes as in the GMT implementation above. Phase mismatch between windows will be proportional to the node distance from source, generally a non-integer period. Event windows are then to be referenced sequentially relative to the machine reset and not to the PPS.

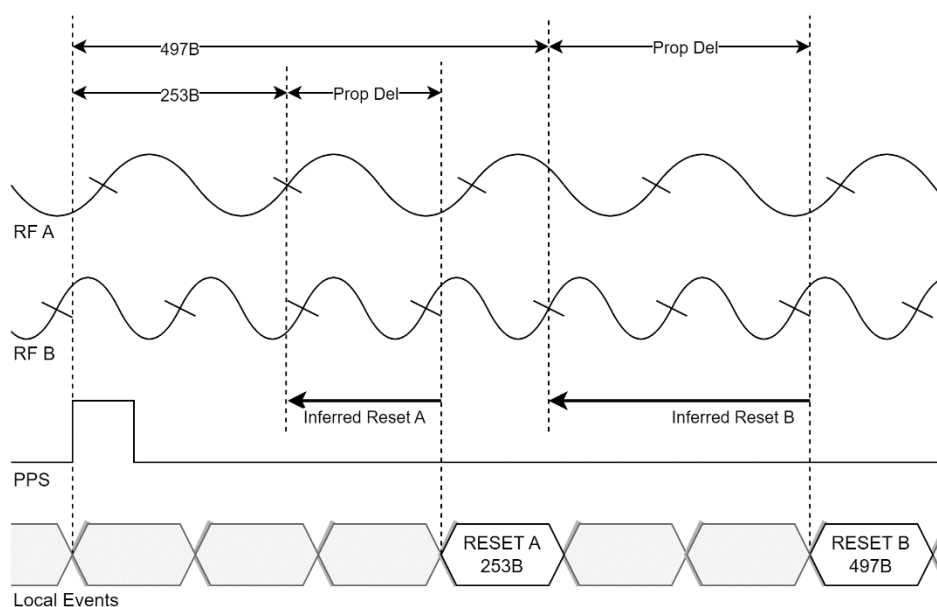
To hold inter-bucket precision with GMT hardware, the reset must be known relative to the PPS. This may be accomplished with the same mechanism as the GMT -> BS proposal above, but events are to be played out in Event Windows with a period proportional to the LLRF. Edge cases (RF jitter across PPS boundaries) will be prevalent. Mitigation of this effect for low frequency sweep machines may be accomplished with local delay lines. A mitigation mechanism for machines where the phase delay may move in excess of one bucket across the PPS boundary between ramps is still under investigation.



4.1 Multi-Machine BS Timing

Devices connected to two or more LLRF networks may track multiple machines at an inter-bunch level by incorporating a 'clock domain' specifier with the GMT data packet. This may be useful for inter-machine transfers, as events can be orchestrated to the bucket level from a single node on the GMT network.

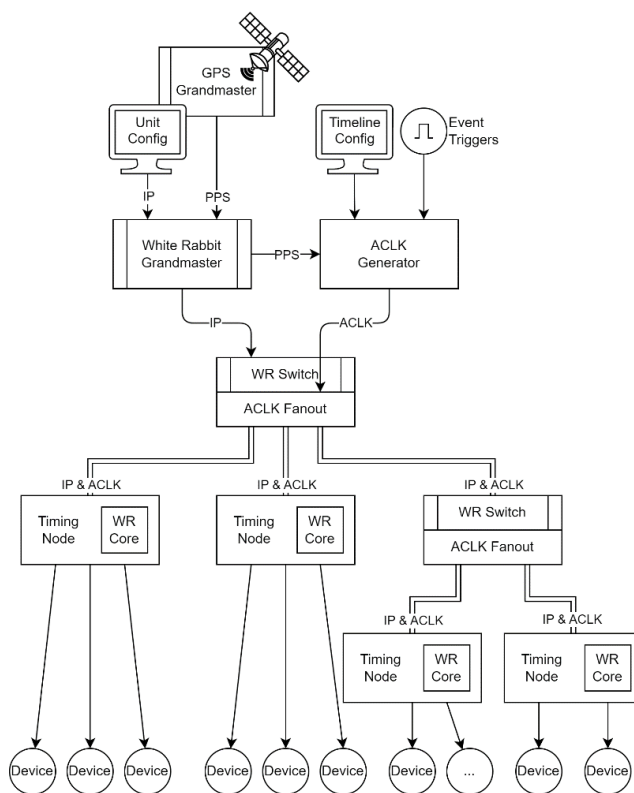
In this scenario, two counters are maintained to track buckets since PPS as in the BS timing examples above. When a reset event is received, it is compared to the specified counter in order to infer the reset and act on subsequent bunches. Error in this scenario will be compounded by the intersection of all 3 clock domains. Exact compensation mechanisms are still under analysis, and this method should not be relied on for exact bunch-to-bunch tracking between RF domains.



Beam Synchronous Timing Across Clock Domains

5.0 WHITE RABBIT INTEGRATION

To address the intrinsic compound errors with real-time clock distribution, local oscillator synchronization via the White Rabbit network is proposed. As outlined above, events are distributed in real time via a dedicated link, and the White Rabbit IP network is utilized for local synchronization and node configuration as shown below. For each device on the timing network, both an Accelerator Clock (ACLK) and White Rabbit (WR) link must be provided. While the parallel WR link appears redundant, it provides already essential Internet Protocol (IP) configuration capabilities for the timing units, as well as an established off-the-shelf solution for time distribution over existing infrastructure. WR links are not required to maintain the precision of short-run timing distribution between nodes and devices.



5.1 Advantages

The hybrid ACLK-WR system proposed provides a high-reliability method for campus-wide event synchronization. It readily enables inter-bucket timing of the PIP-II 162.5MHz bunch structure and has potential to synchronize events at an inter-bucket level between Booster and Main Injector. This system is backwards compatible with existing TCLK infrastructure and may be successfully executed utilizing off the shelf hardware within Fermilab's current engineering capabilities.

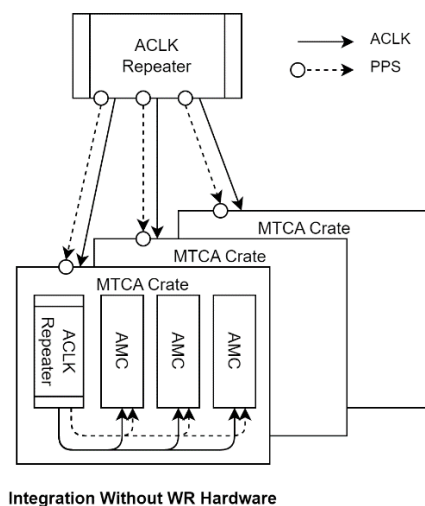
5.2 Integration With External Systems

For devices not connected to both the White Rabbit and ACLK links, integration may be maintained as follows:

5.2.1 PRECISION TIME INTEGRATION THROUGH PPS:

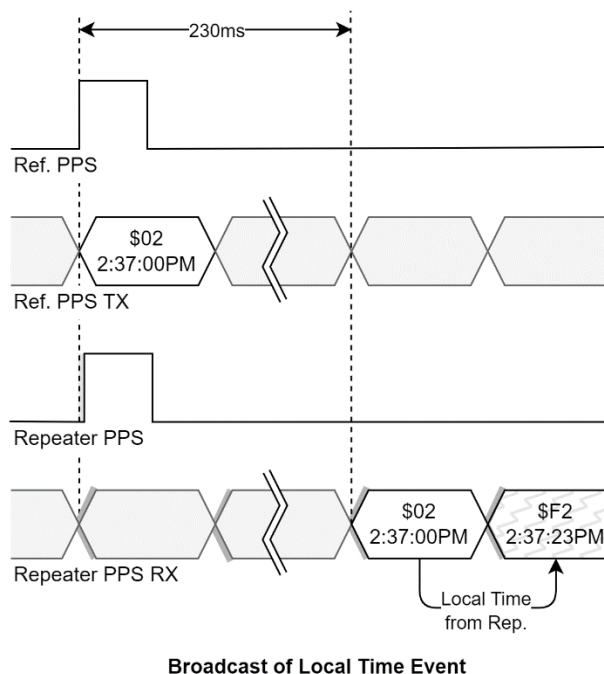
For systems without White Rabbit capabilities, an external Pulse Per Second may be sourced from the WR link of the local ACLK repeater. This PPS signal will present as a TTL pulse run in parallel to the ACLK link and provide a similar level of precision, provided the transmission length is direct and no longer than 100m. Synchronization and decoding may then be performed in the same manner outlined above. Without a local 10MHz oscillator, events should not be re-broadcast outside of the crate.

Here a Micro TCA crate is used as an example. A standard AMC or MCH receives the ACLK and PPS signals, then transmits both over the backplane to provide dedicated links for each slot.



5.2.2 TIME INTEGRATION THROUGH LOCAL SYNCHRONIZATION EVENTS:

For systems requiring lesser degrees of synchronization, a high precision local timestamp event may be generated at the repeater. In this scenario, the ACLK repeater uses the WR-PPS synchronization methods outlined to maintain precise local time. After every 'global' synchronization event, a 'local' time event is generated according to the actual time at the node and broadcast downstream. Without PPS integration, this timestamp may be considered accurate within one granularity window.



6.0 TECHNICAL DETAILS

6.1 Event Types

The following event types are defined. These event types are derived from the 1986 TCLK specification, appended with system specific event types taking advantage of ACLK's extended data frames. The event types are non-comprehensive.

6.1.1 GLOBAL SYNCHRONIZATION EVENT

To hold timing and align nodes to the ACLK network, a priority zero, Global Synchronization Event (GSE) shall be broadcast at one second intervals in conjunction with the PPS from the White Rabbit grandmaster. The GSE shall have a clock domain of zero and contain the 64b epoch time (in nanoseconds) of the immediate PPS (fractional seconds should be zero). The intent of this event is to provide a timing reference for downstream nodes relative to the exact network conditions.

6.1.2 BEAM REQUEST EVENTS

For each downstream machine involved with beam transfer, a Beam Request Event (BRE) shall be available (not necessarily implemented), and broadcast with high priority. The BRE shall contain the Source Machine ID (SMID), Target Machine ID (TMID), and the anticipated bucket count (or time) at which beam may be accepted by the downstream machine. The intent of the BRE event is to provide a mechanism for beam transfer over general ACLK infrastructure, and/or method for a given machine to self-verify local conditions and permits.

6.1.3 TRANSFER ID (PULSE ID)

For each machine transfer, a preceding Transfer ID (TID) shall be broadcast with medium priority before any Transfer Events are executed. The TID shall contain a SMID, TMID, and a 64b Pulse Identifier (PID), as assigned by the Beam Switch Sum Box (BSSB). The PID shall be a unique number, incrementing sequentially with each transfer from the machine. The PID shall have a corresponding truncated identifier containing the last 8b of the PID (PID8) for general use. The intent of the TID event is to definitively correlate machine cycles for logging and control purposes.

6.1.4 TRANSFER PRE-TRIGGER

For each machine involved with beam transfer, a Transfer Pre-Trigger (TPT) event shall be broadcast by the extraction system with very high priority at the earliest moment at which the transfer conditions are known with high certainty. This TPT shall contain the SMID, PID8, and the predicted bucket count (or time) at which the extraction is to occur. The intent of the TPT event is to allow downstream systems adequate time to prepare, sample, or act on synchronously with the incoming bunch train.

6.1.5 TRANSFER CONFIRMATION

For each machine involved with beam transfer, a Transfer Confirmation Event (TCE) shall be broadcast with very high priority after the transfer sequence has been initiated by the extraction system. The TCE shall contain the SMID, PID8, and the actual bucket count (or time) at which the extraction occurred. The intent of the TCE is to allow post-transfer correlation, and corroboration of the estimated trigger point with the actual time of trigger.

6.1.6 TRANSFER STATUS

For each machine involved with beam transfer, a Transfer Status Event (TSE) with medium priority shall be broadcast after all transfer events are executed. The TSE shall contain the SMID, PID, and a data field with beam charge and energy at time of transfer.

6.1.7 INTER-CYCLE-TRIGGER

For each machine involved with beam storage, an Inter-Cycle-Trigger (ICT) shall be available with very high priority for actions to be executed at an exact point in the beam cycle. The ICT shall contain the SMID, TMID, and the bucket count (or time) at which the action is to be taken. The intent of the ICT is to allow flexible, inter-cycle actions to be taken in real time.

6.1.8 MACHINE RESET

For each machine, a Machine Reset (presently just Reset) event shall be available with medium priority. These events are generated by the Timeline Generator (TLG) in order to perform general orchestration and clearing of machine status. The reset shall contain the SMID and TMID. The data field for the reset event is not yet defined.

6.1.9 ABORT EVENTS

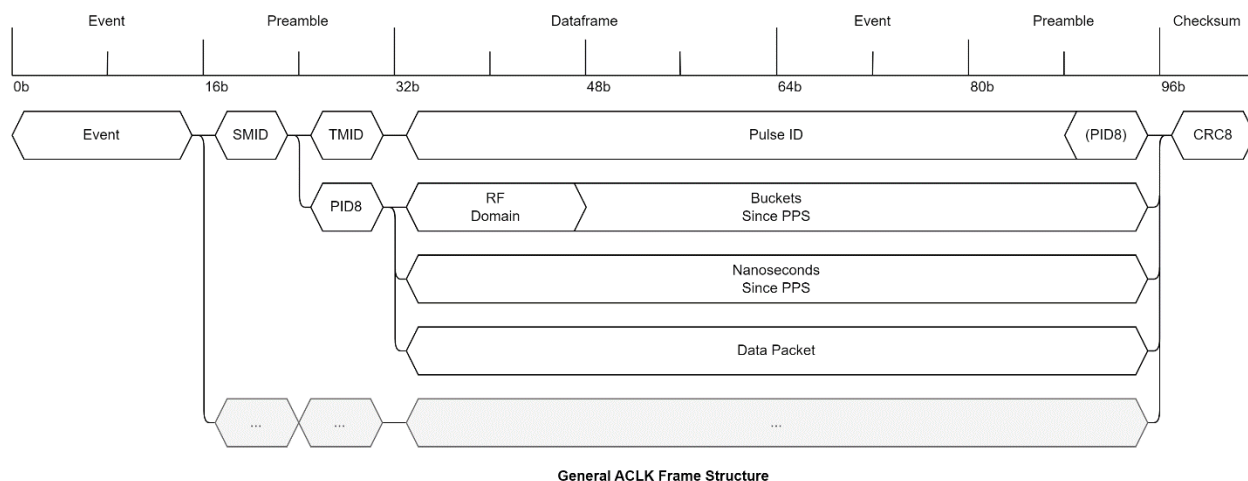
Abort events shall be made available with high priority for all systems connected to the ACLK network. The abort event shall contain the SMID, PID8, and associated timestamp of the abort.

6.1.10 PERIODIC TIMING MARKERS

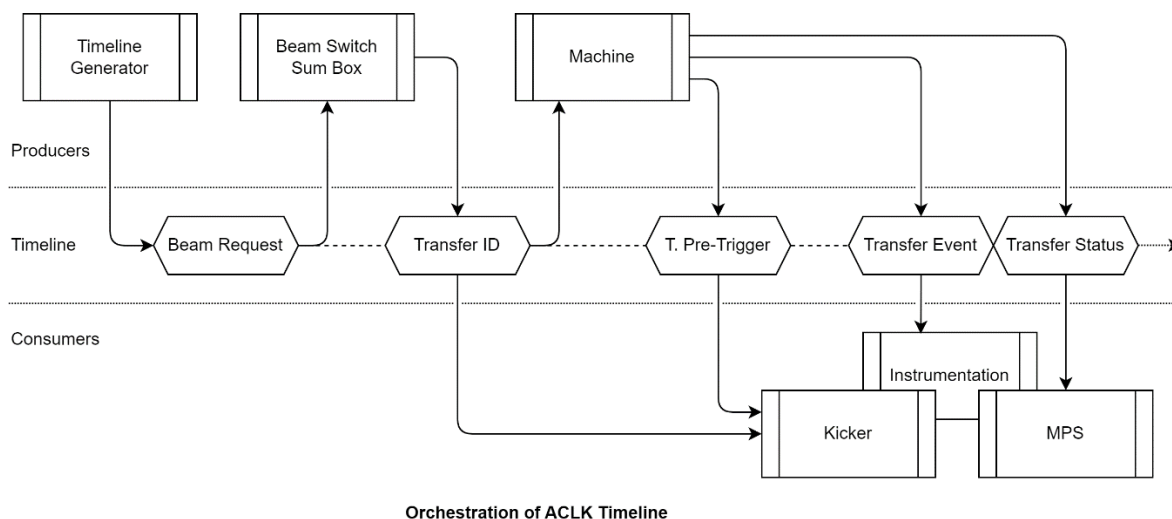
To maintain timing for systems dependent on general machine harmonics (20Hz, 60Hz, 720Hz) or periodic events (Super Cycle Reset), lower priority synchronization events shall be broadcast periodically with a jitter no greater than 20uS.

6.2 Frame Construction

Below is a proposed frame construction method for encoding multiple event types into a uniform protocol. While some serial protocols do not necessarily require frames of fixed length, utilization of a uniform frame protocol simplifies timeline orchestration logic in ensuring frames are deterministically sequenced and passed in a fixed amount of time.



The timeline may be constructed as follows. Here ‘producers’ (TLG, BSSB, and Machine) insert events into the timeline via direct data links with the ACLK generator. These are broadcast to ACLK global, and acted upon by downstream systems in real time.



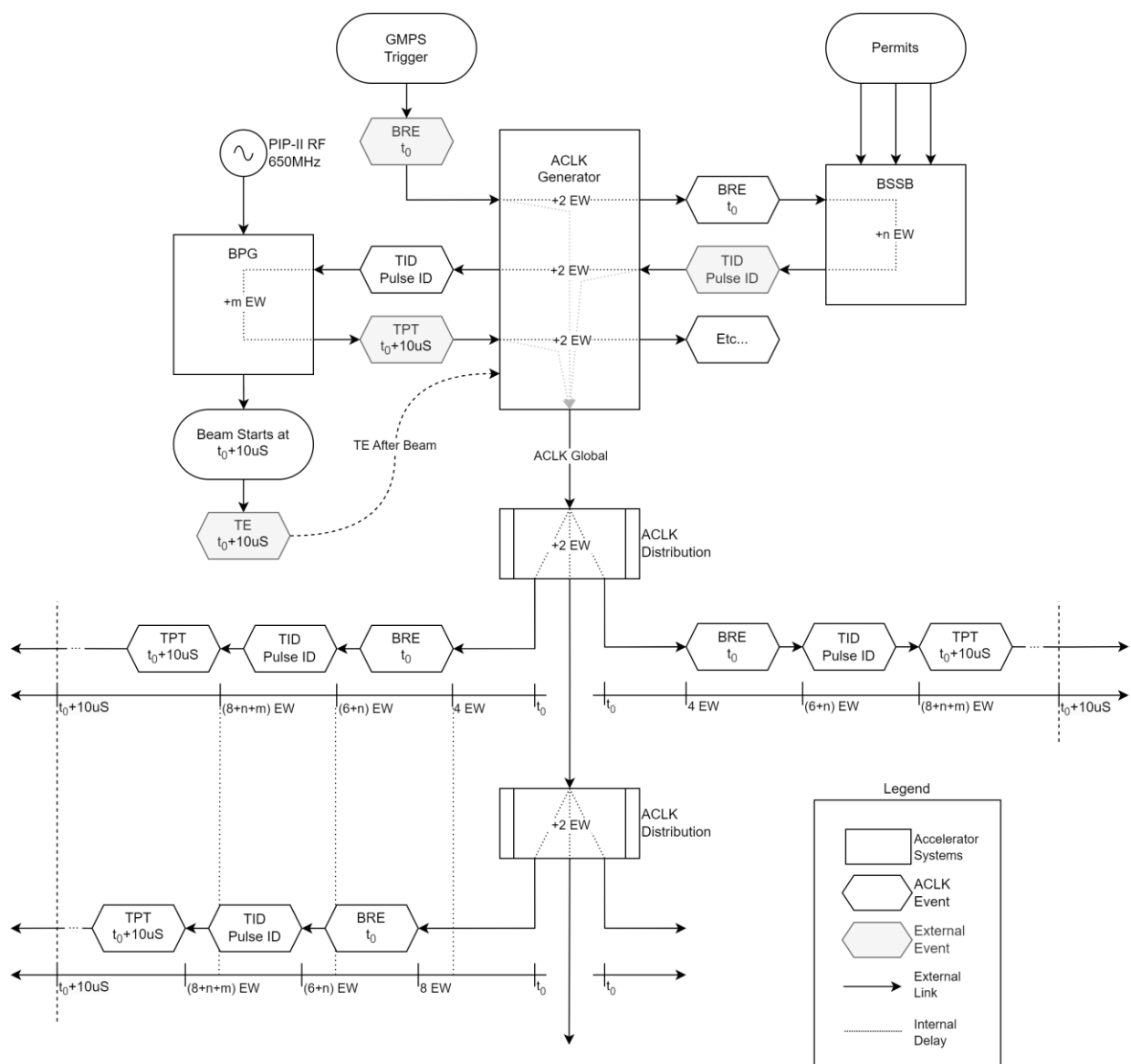
7.0 USE CASES

The following scenarios are identified.

7.1 Example Timeline Generation for PIP-II

Given the RF synchronous example above, a precision real-time system capable of inter-bucket discrimination may be constructed. In this scenario, precision timestamps and/or data are encoded within the ACLK triggers by upstream devices. The event/data packet pairs are re-broadcast at corresponding event windows as outlined. The order of operations is as follows:

1. The Gradient Magnet Power Supply (GMPS) determines the start of Booster cycle, and sends a Beam Request Event (BRE) with the target time to the ACLK Generator
2. The ACLK Generator broadcasts the BRE to the Beam Switch Sum box (BSSB), which validates the permits are in place and responds with a Transfer ID (TID) event.
3. The ACLK Generator broadcasts the TID to the Beam Pattern Generator (BPG), which synchronizes the local RF, and responds with the Transfer Pre-Trigger (TPT) event containing the time (or bucket) at which the beam pattern will play out.
Note! This time must be sufficiently in advanced (~10uS, 6.5K RF cycles) to accommodate the propagation delay requirements of downstream systems.
4. The ACLK Generator concurrently broadcasts all events to downstream distribution devices as received, allowing nodes to arm and act on the beam in real time.
5. At the pre-determined time, the beam is generated. The BPG sends subsequent Transfer Confirmation Event (TCE) containing the exact time (or bucket) at which beam was initiated and subsequent Transfer Status Event (TSE) information.

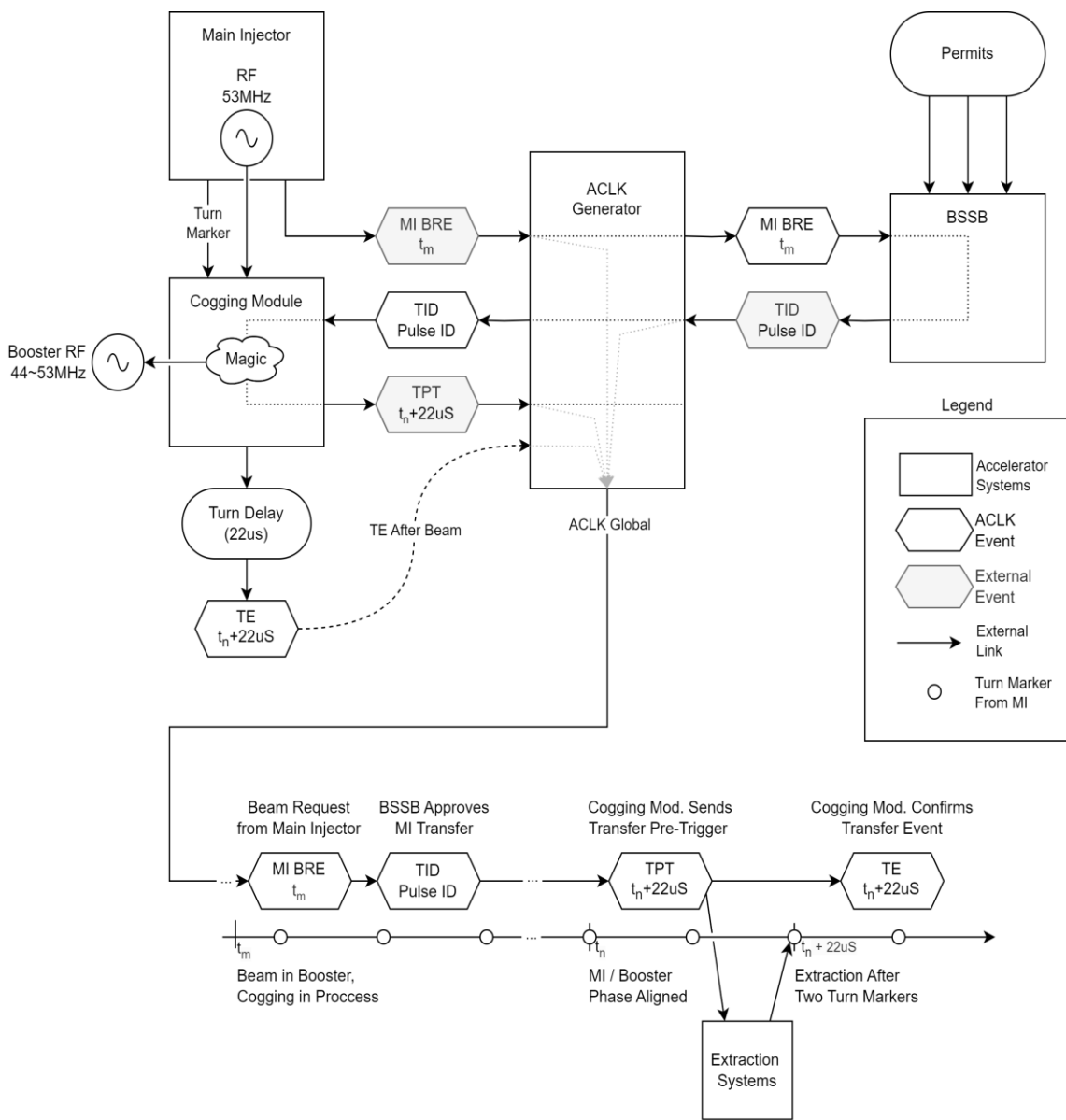


7.2 Example Timeline for Booster Cogging

Given the RF synchronous example above, Booster Cogging (transfer) events may be orchestrated to a high degree of precision. At Fermilab, the Booster is responsible for phase-aligning RF buckets between the PIP-II Linac and Main Injector through the transition period. This operation is an active feedback system, with possibility for variable time or bucket delays on a cycle-to-cycle basis.

For this scenario, the Cogging Module will broadcast Transfer-Pre-Trigger events after alignment with Main Injector is known, but before extraction is to occur. This TPT allows for downstream systems to track buckets and execute in a synchronous manner on event. This is a similar concept to the Tawzer Modules presently used for extraction systems (counting buckets and applying a vernier delay) but acts on a global scale. Assuming relevant systems are bucket aligned, order of operations is as follows:

1. The Main Injector sends a general Beam Request Event with target time of transfer to the ACLK Generator. This is validated by the BSSB, and assigned a Transfer ID.
2. The cogging module receives this BRE and TID during the cogging cycle. After phase alignment with Main Injector, it broadcasts the Transfer Pre-Trigger, with a time or bucket offset at which extraction is to occur (in this scenario 22uS)
3. The Cogging module holds this phase alignment for a sufficient number of cycles for the event propagation delay to reach downstream systems, as in the PIP-II example above.
4. At the pre-determined time, extraction systems act synchronously on this transfer event.
5. The Cogging Module or extraction system sends subsequent Transfer Confirmation Event (TCE) containing the exact time (or bucket) at which transfer was initiated.



Example Booster Timeline, With Extraction Pre-Trigger

8.0 APPENDIX A: CONCEPTS

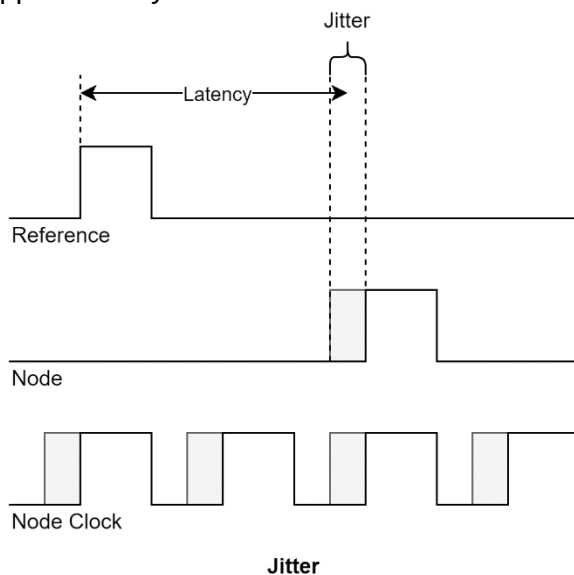
The following items have been explicitly defined in order to assist the conversation:

8.1 Latency

Latency is the time it takes for a signal to propagate from point A to point B. This term may also be referred to as Propagation Delay. The singular term 'Delay' should be reserved for intentional offsets of a clock either in hardware or by length matching, not the fundamental propagation delays between nodes in a network. Static latencies may be easily corrected for in timing networks. The nominal latency of copper and fiber is approximately 5uS/Km.

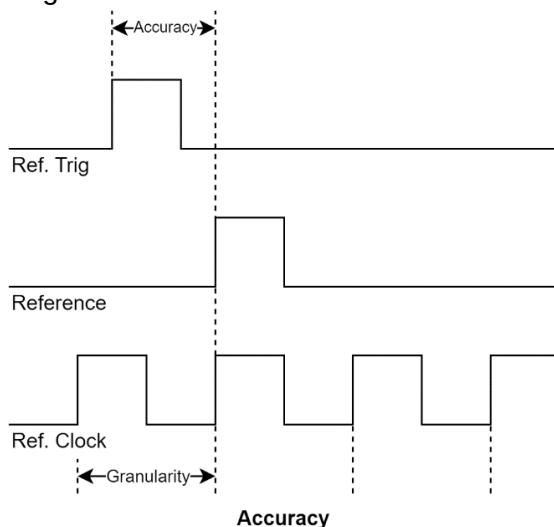
8.2 Jitter

Jitter is the *unintentional* statistical deviation in latency around an event edge due to a variety of external factors. Jitter may be imparted to an edge by changes in propagation delay (cable length changes with temperature), repeater configurations, or crossing of clock domains (see accuracy). Jitter correction requires active feedback as network conditions change with time. The nominal jitter for copper is approximately 7nS/Km over a 25°C temperature range. The nominal jitter for fiber is approximately 1nS/Km.



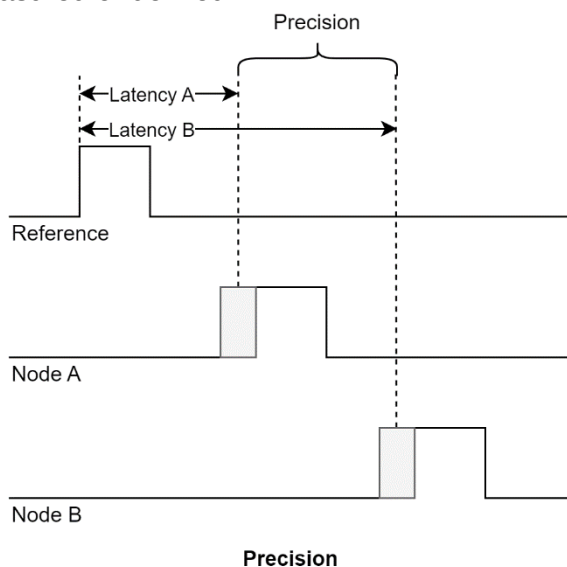
8.3 Accuracy

Accuracy is the deviation of event edges when compared between the originating 'event' and time at which it was encoded into the timing network. Accuracy is foremost determined by the period, or 'Granularity' of the timing network clock domain. Accuracy of downstream nodes may also be impacted by jitter. Accuracy is not a function of latency, as this offset is static. Accuracy for the present Fermilab timing network has not been measured or defined.



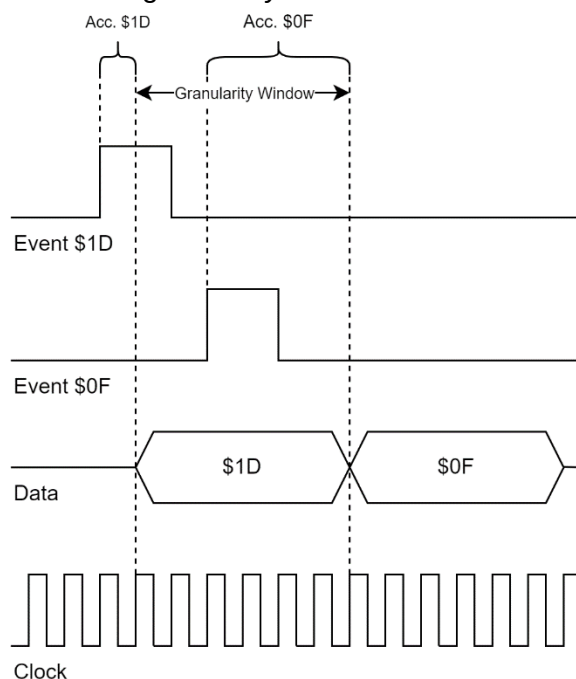
8.4 Precision

Precision is the deviation of event edges when compared between any two nodes or two periodic events measured at the same node in a timing network. Precision is not a function of latency, as this offset is static. Precision is instead a compounded function of jitter between the edges (gray box + gray box in the diagram below). Precision for the present Fermilab timing network has not been measured or defined.



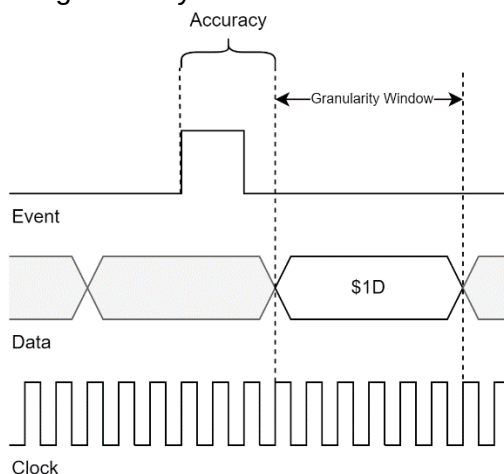
8.5 Granularity Window

The Granularity Window is the compounded period (multiple clock cycles) during which events are played out. For a single event, the granularity may be as low as one clock cycle. When multiple events are inserted within the span of a single event's encoding window however, the accuracy of subsequent events will be negatively impacted. The 10Mbaud TCLK protocol with 12b event definitions has a variable granularity window between 100~1200nS.



Variable Granularity Window

For timing networks with fixed frame definitions (events may only be distributed every 'n' clock cycle) the granularity window will be fixed at the encoding time ('n' * the clock period). The 650Mbaud ACLK protocol with 65b event definitions has a fixed granularity window of 100nS. This document assumes fixed granularity windows for most discussion.



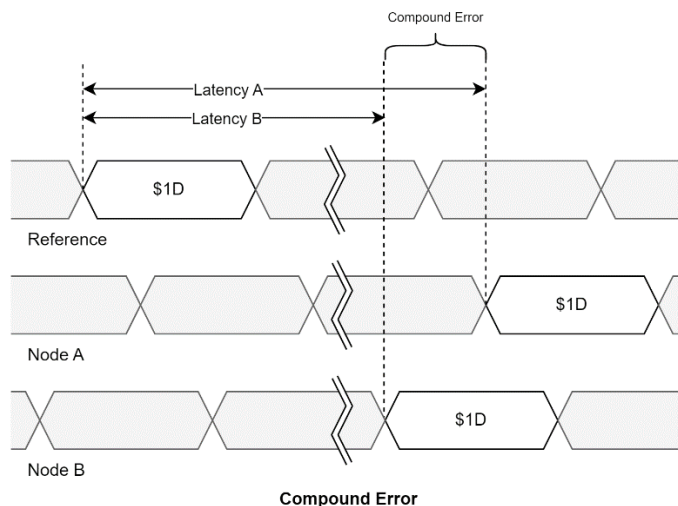
Fixed Granularity Window

8.5.1.1 A TECHNICAL NOTE ON THE WHITE RABBIT GRANULARITY WINDOW.

The White Rabbit protocol is an ethernet framework that synchronizes clock edges across large area networks with a high degree of accuracy and precision ($< \text{nS}$ between clock edges). Some applications also use this ethernet protocol for event distribution. Due to inherent latencies in IP communication however, the granularity window is typically locked to 100uS frames (with the possibility of executing multiple events within a frame). This approach is sufficient for modern accelerator systems, but the unique characteristics of Fermilab's Booster complex may see changing conditions within the 100uS window that require different timing of subsequent frames. For this reason White Rabbit is not a viable option for timing system event distribution at Fermilab, but its jitter correction methods may still be applicable.

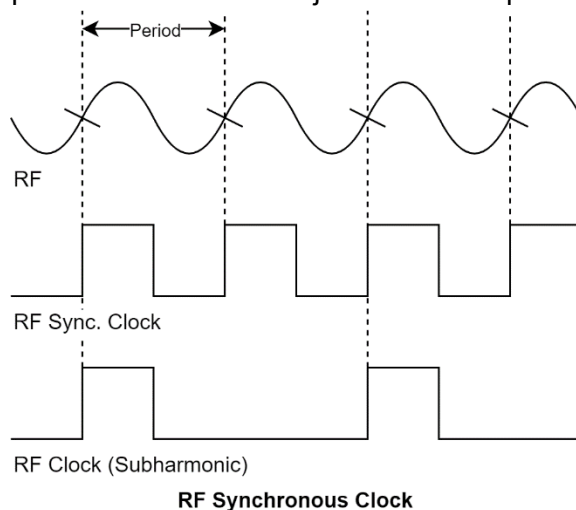
8.6 Compound Error

Event jitter between nodes is considered a compounding error. For a machine timing system to work reliably, it must be guaranteed that all events are executed within the same granularity window (irrespective of latency). This means the compound error must be significantly less than the granularity window for every node on the timing system network, or events may periodically slip into adjacent windows. The impact of this slip will vary by event, but in general should meet the tightest timing requirement.



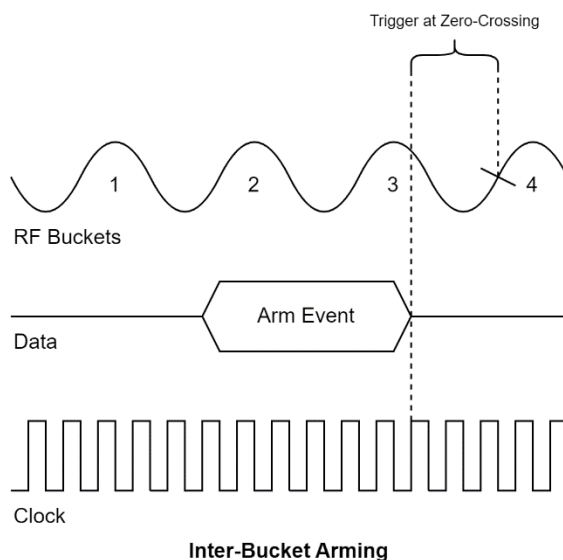
8.7 RF Synchronous Clocks

For precision timing applications it may be advantageous to lock the clock frequency to the frequency of the machine. This approach ensures that events will always arrive at the desired bucket. RF synchronous clocks may be a multiple of the machine RF, or a sub-harmonic of the machine RF. The 650Mbaud LCLK protocol is 4x PIP-II's RF frequency of 162.5MHz. The 7.5Mbaud MI Beam Sync protocol is $1/7^{\text{th}}$ Main Injector's RF frequency of 53 MHz.



8.8 Arming Events

For precision RF systems synchronized to the general timing network (kickers, laser wires, BPM's), clock events (or discrete triggers) and local RF frequencies may be utilized to 'arm' a system, then act on a given bucket. In this instance, the arm event arrives at a known offset before the target bucket and the system counts zero crossing events before firing in sync with the RF. Less precise arming events may also be used to capture baseline signal preceding a beam event.



8.9 Event Jitter

From the factors listed above, event jitter is of major concern for RF synchronous actions. Clearly defined tolerances are required to assign priority and maintain the reliability of precision events. As shown by the example below, event jitter in excess of one RF period (6~20nS) is sufficient to break inter-bucket timing. For this reason, jitter should be tightly controlled when executing RF-synchronous events from general purpose clocks.

